DUT RES MANAGEMENT SYSTEM

Group 51

FACULTY OF ACCOUNTING AND INFORMATICS

DEPARTMENT OF INFORMATION TECHNOLOGY

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4. We have not used other students' past work to hand in as our own.
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Please complete this form and submit it as a cover sheet every time you submit your project. One submission is required per group. The details completed below must be correct.

Student No.	Surname, Initials	Group (App Dev, App Dev 4Yrs, or BA)	Task Completed	Signature
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Proposal for DUT RES MANAGEMENT SYSTEM

1. BUSINESS DESCRIPTION/OVERVIEW

DUT Housing department plays a crucial and important role in ensuring that student living is simple, safe, supportive and convenient. With that being said, the housing department relies a lot on manual operations such as keeping student records in books as well as all of the complaints that arises from the residence, this causes a lot of inconveniences and a lot of time gets wasted.

This proposal's aim is to introduce a Digital Res Management system that promotes green computing and saves a lot of time by keeping all records digitally in a database and allocate students automatically to their rooms as they check in at the residence, it also has added features which would be keeping track of visitors in the residence and allowing students to post maintenance issues within the residence. This will make it easy for staff to keep track off all issues that are pending and those that have been resolved.

2. CURRENT SYSTEM/BUSINESS PROCESSES

APPLICATION & ALLOCATION PHASE

1. Step 1: Emails are sent to students inviting them to apply for residence.
2. Step 2: Housing receives a code indicating whether a student is allocated to a residence (res) or placed on the Pending Allocation (PA) list.
3. Step 3: If coded as res, students are allocated to residences by faculty, if coded as PA, allocation is done randomly.
4. Step 4: Students are registered to their allocated residence.
5. Step 5: Housing issues a proof of residence document.

Room Allocation Phase

6. Step 6: Students go to the residence, queue, and submit a hard copy of their proof of residence for verification.
7. Step 7: Rooms are allocated on a first-come, first-served basis.
8. Step 8: If the residence is full, students are redirected to Housing to apply for a different residence.

House Committee Election Phase

9. Step 9: The year before, Housing issues a link for House Committee applications.
10. Step 10: Applicants must have a 65% aggregate; those who don't meet this are disqualified.
11. Step 11: A list of qualifying applicants is published for seconding.
12. Step 12: A date is set for elections, and results determine the Chairperson, Secretary, and Project Officer.

Residence Advisor (RA) Selection Phase

13. Step 13: Applications for RA positions are opened.
14. Step 14: Applicants must:
    - Hold a qualification.
    - Have 65% in that qualification.
    - Be pursuing a Master's or advanced degree.
    - Those who don't meet these criteria are disqualified.
15. Step 15: Qualified applicants are interviewed, with preference given to former DUT leaders.
16. Step 16: RAs are elected and allocated to residences by faculty.
17. Step 17: Housing sets a date for portfolio training for elected RAs.

Residence Operations & Maintenance Phase

18. Step 18: A WhatsApp group is created by the House Committee and RA for all students in the residence.
    All maintenance issues are reported in this group.
19. Step 19: The RA reports issues to the Building Manager, who then reports to the Landlord.
20. Step 20: A maintenance book is kept and checked weekly by the Building Manager.

3. PROBLEM DESCRIPTION/CHALLENGES

DUT Housing Department ensures that students are allocated to their residences, resolve issues that may arise, they also ensure that student comply with their policies. Current processes include:

- Manual Applications: Students apply for residences through email and are allocated based on faculty or random assignment.
- Physical Room Allocation: Students stand in long queues and submit hard copies of proof of residence for verification.
- Maintenance Reporting: Issues are reported in WhatsApp groups or handwritten logbooks, which are reviewed only weekly.
- Visitor Registration: Visitors are registered manually in a physical book, making tracking inefficient.

Key Challenges:

- Long queues and delays in room allocation.
- Risk of data loss or misinterpretation due to hard copy records.

- Slow response to maintenance issues as logs are checked weekly.
- Lack of real-time monitoring of visitors and residence occupancy.

4. PROPOSED SOLUTION AND SYSTEM CAPABILITIES

The proposed solution is a web-based and mobile-friendly system that digitizes and integrates all core residence management activities into a single platform.

System Capabilities:

1. Student Records Management

- Stores student credentials and residence allocation details in a secure database.
- Provides real-time access to records for administrators.

2. Online Application & Room Allocation

- Students apply online for residence.
- students are allocated automatically based on availability, faculty, and preference criteria.
- Eliminates physical queues through digital room allocation.

3. Maintenance Management

- Allows students to report maintenance issues via the platform.
- Tracks resolution progress and timelines for quick response.

4. Visitor Management

- Digital registration of visitors with ID verification(Student number, ID number).
- Tracks visitor entry and exit times to ensure visiting policies are not violated.

5. House Committee and RA Management

- Digital platform for house committee applications, elections, and role allocation.
- Integration of announcements and communication tools for committees.

6. Green Computing & Data Security

- Reduces paper usage through complete digitization.
- Implements encryption and secure authentication for data protection.

ENTITIES

1. Institution - Represents DUT and its housing department, overseeing all residence operations.

2. Student - Contains student credentials, residence status, complaints, and room allocation.

3. Visitors - Contains visitor credentials, check-in and check-out times.

4. Maintenance - Logs reported issues, status updates, and resolution timelines.

5. Admin - Housing staff responsible for managing allocations, complaints, and oversight.

6. Room - Tracks room status, occupancy, and allocation history.

7. Residence - Includes residence details, room availability, and assigned students.

8. Security - Verifies identity and monitors access to residences.

5. BENEFITS

1. Efficiency: Eliminates manual processes, saving time for students and staff.
2. Data Security: Protects student records through secure digital storage.
3. Improved Communication: Real time updates on maintenance and housing notices.
4. Sustainability: Reduces paper use and promotes eco-friendly practices.
5. Compliance: Enforces visitor and residence policies through automated checks.
6. Promotes green computing and saves money: By making all processes digital a lot of money for buying books is saved and less paper required.

6. TECHNOLOGY STACK

Component	Technology Used
Mobile App	Java (Android Studio)
Web Dashboard	ASP.NET MVC (C#)
Database	SQL Server
API/Backend	ASP.NET Web API
Authentication	DUT Student Email Login / Firebase Auth / Biometric (Optional)
UI/UX Design	XML for Android UI, Razor Views for Web

7. IMPLEMENTATION PLAN

Phase 1: Information Gathering & System Design

- Conduct detailed analysis with the housing department and collect information.
- Develop system plan and design mockup.

Phase 2: Database Integration

- Build central database that stores student registration, room allocation, maintenance, and visitor records.

Phase 3: Testing & Pilot Deployment

- Test system with a single selected residence.
- Collect feedback and optimize performance.

Phase 4: Full Deployment & Training

- Deploy system to the selected number of residences.
- Create a manual guide that explains how to use the system.

- Host training sessions with staff and students to show them how to use the system.

8. CONCLUSIN

The propose System will be of great help to the Housing department since it will ensure that a lot of money is by reducing the number of books the company buys since all records will be stored online, this will also promote green computing. Not only will it save money but it will also save time since nearly all processes will be automated.